Tobias Mwangi-Hartley
Application: Machine Learning Engineer - Customer Support AI
Quillstone Payments

Dear Hiring Team,

At Thistlebox, I build machine learning models used in weekly operational decisions, from recipe demand forecasts to retention targeting. I am applying for the Machine Learning Engineer - Customer Support AI role to bring that experience in Python, feature engineering and model evaluation to your AI team while developing deeper production engineering expertise. Your team's work with backend engineers, data scientists and product managers to take ideas from prototype to production aligns with this next step.

My LightGBM forecasting model predicts orders six weeks ahead and reduced WAPE from 23% to 14%, with an estimated 9% reduction in food waste. It runs as a Vertex AI Workbench scheduled notebook, writes to BigQuery and supplies forecasts that operations accesses through Looker. I also developed an XGBoost churn model scored weekly in BigQuery; validation using a 10% holdout showed churn 2.1 percentage points lower in the treated group. These projects provide a practical foundation in feature engineering, recurring data workflows and evaluating business outcomes.

I have also begun strengthening the engineering around my work. Since 2025, I have added pytest unit tests to forecasting feature code, reaching approximately 60% coverage. For a sequential A/B testing calculator used by approximately 25 colleagues, I wrote the Streamlit application and Dockerfile, while the data platform team handled deployment to Cloud Run. My experience to date is in scheduled and batch workflows rather than API model serving. Alongside this, my MSc in Statistics and three deliveries of "Stats for PMs" support rigorous evaluation and clear communication with non-specialist colleagues.

I would bring this combination of applied modelling, measurable operational results and growing engineering practice to your Customer Operations AI team. I would welcome a conversation about how my experience could contribute to systems that help merchants get support quickly and without friction. Thank you for considering my application.

Kind regards,
Tobias Mwangi-Hartley